Red Pawn

MP3: Alice Factions, Behavior, and Counterplay · Individual Submission

Jaeshin Jung · VIST 386 · Red Kingdom faction · Unit: Pawn

1. Unit Identity and Role

Who it is. The Red Pawn is the rank-and-file spear soldier of the Red Kingdom, drawn from the chessboard of *Through the Looking-Glass*. In the book, Alice begins as a pawn and becomes a queen by reaching the far side of the board. The Red Pawns are the soldiers standing in her way.

Function in the faction. The Pawn is the faction's front line and its eyes. It screens for the Queen, Rook, and Knight, and it spots the player so the stronger units can react.

What makes it different. The Pawn is the only unit whose power comes from how many of its own kind are nearby. Alone it is the weakest unit in the faction. In groups it gains three abilities: Phalanx, Shared Sight, and Coronation (a group promotion into a temporary Knight, Rook, or Queen). The Queen, Rook, and Knight are individually strong, while the Pawn is strong only as a crowd. The chess source shapes behavior, not just looks: Pawns march forward in ranks, and promotion, the pawn's defining chess rule, becomes the group's escalation.

2. Threat, Power, and Counterplay

What makes it threatening

- **Moving jab:** a short-range forward thrust while walking.
- **Braced thrust:** when the Pawn stops and plants its spear, it attacks from much farther away after a visible wind-up.
- **Phalanx:** a braced Pawn can thrust past an allied Pawn standing in front of it, so a line of Pawns stacks attacks in one direction.
- **Shared Sight:** if one Pawn spots the player, nearby Pawns learn the position.
- **Coronation:** when a group of three or more Pawns loses one, the others promote the front-most Pawn into a temporary Knight, Rook, or Queen.

Power curve position. Alone, the Pawn is the lowest tier in the Red Kingdom. With three or more Pawns it becomes a mid-tier threat because of overlapping thrusts and shared information. Coronation causes a short spike in power during the fight. Promoted forms stay low on HP and last only a limited time, so they never exceed the real Queen, Rook, or Knight. Numbers in this document (ranges, durations, odds) are starting values to tune in playtests.

Counterplay

- Kill the Pawn that spotted you before the others learn your position.
- Attack from the side. The Phalanx only reaches forward.
- Fight moving Pawns up close, since their moving jab has short range.
- Bait a Pawn into bracing, then step out of its line. A braced Pawn cannot move.
- Interrupt Coronation by killing the chosen Pawn during the ceremony, when it is immobile and vulnerable.
- Break up groups before they reach three Pawns.

Weaknesses and limitations. Each Pawn has a narrow, short view, almost no coverage to the sides or rear, and cannot move while braced. Small groups cannot trigger Coronation.

3. Information Model

Question	Pawn
What it perceives	A narrow, short forward cone. When one Pawn sees the player, nearby Pawns receive the player's position (Shared Sight).
What it does not know	Anything behind or beside it, anything outside its cone, and the player's true position once sight is lost. It does not know the player's plans.
Memory	It remembers the last known player position for a short time (a few seconds, tunable), then forgets it and resumes marching.
Can the player hide or deceive it?	Yes. The player can break line of sight, stay outside the cone, leave the group chasing a stale last known position, or kill the spotter before it shares the position.

4. HSM Diagram (Paper Design)

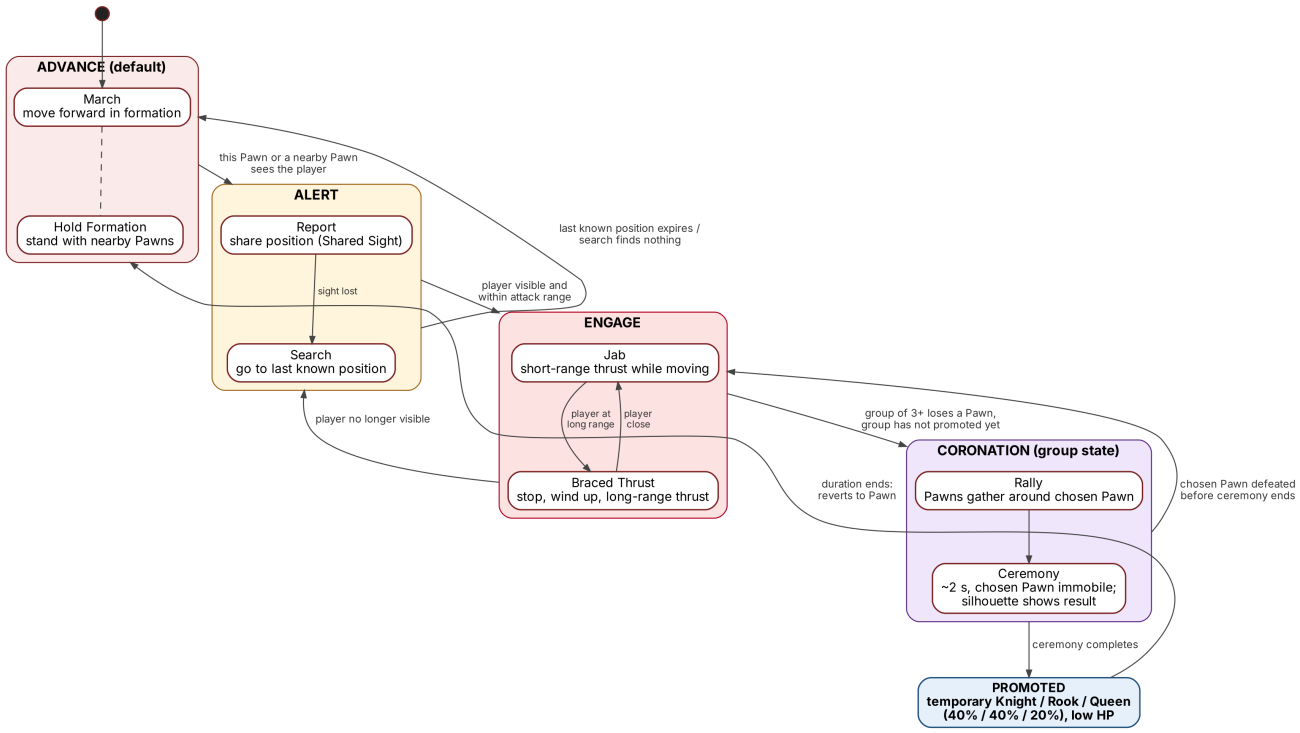


Figure 1. Pawn Hierarchical State Machine. Shaded boxes are composite states with substates; arrows are labeled with the condition or event that causes the transition.

From	To	Condition or event
ADVANCE	ALERT	This Pawn or a nearby Pawn sees the player (Shared Sight)
ALERT	ENGAGE	The player is visible and within attack range
ALERT	ADVANCE	Last known position expires or Search finds nothing
ENGAGE	ALERT	The player is no longer visible
ENGAGE	CORONATION	Group has 3+ Pawns, one Pawn is defeated, and this group has not promoted yet
CORONATION	ENGAGE	The chosen Pawn is defeated before the ceremony ends
CORONATION	PROMOTED	The ceremony completes
PROMOTED	ADVANCE	The promoted form's duration ends and it reverts to a Pawn

5. Behavior Tree (Paper Design): ENGAGE State

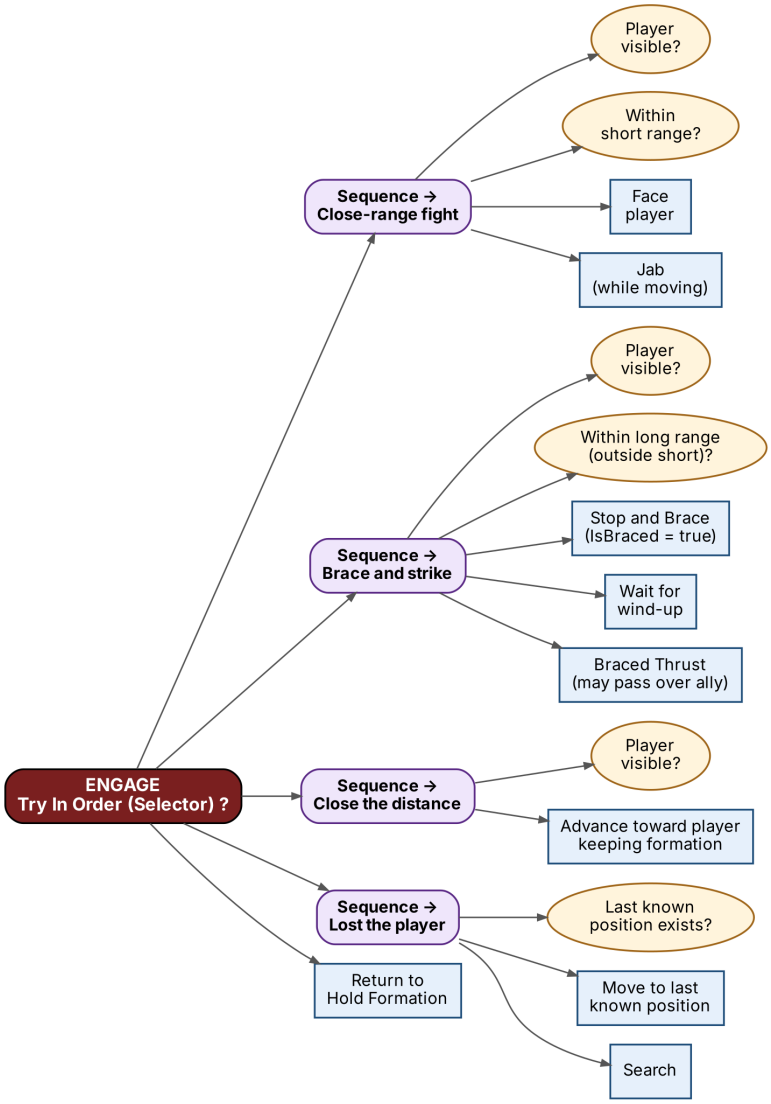


Figure 2. Behavior Tree inside ENGAGE. Purple = control nodes (Try In Order selector, Sequences), yellow = conditions, blue = actions. Children run top to bottom in priority order.

The tree has five control nodes (one Try In Order and four Sequences), more than six condition and action nodes, and genuinely different behaviors: a moving Jab, a stationary Braced Thrust, advancing in formation, and searching the last known position. It decides *how* to fight based on distance, which is the decision implemented in the Unity proof (Section 10).

6. Blackboard Variables

Name	Type	What it represents	What can change it
PlayerVisible	bool	Whether this Pawn currently sees the player	Vision cone check
LastKnownPosition	Vector3	Where the Pawn last saw the player, or was told the player is. Memory, not the player's real position.	Updated when the Pawn sees the player or an ally shares a sighting; expires after a short time
AllyReportedPlayer	bool	A nearby Pawn shared a sighting this Pawn did not make itself (secondhand, imperfect information)	Shared Sight message from a nearby Pawn
PlayerDistance	float	Distance to the player, valid only while the player is visible	Distance check
NearbyAllies	int	Number of Pawns close to this one	Proximity check
IsBraced	bool	Whether the Pawn is planted for a long thrust	Brace and Jab actions
CurrentMode	enum (Advance, Alert, Engage, Coronation, Promoted)	The high-level HSM state	State transitions

LastKnownPosition and **AllyReportedPlayer** are the imperfect-information and memory variables: they hold what the Pawn believes, not where the player actually is.

7. MDA+A Analysis

Mechanics

- Moving jab (short range) versus braced thrust (long range, immobile, visible wind-up).
- Phalanx: a braced Pawn can thrust over an allied Pawn in front of it.
- Shared Sight with a narrow individual view and short memory.
- Coronation: triggered when a group of three or more loses a Pawn, once per group, with a ceremony of about two seconds and a weighted random result (Knight 40%, Rook 40%, Queen 20%).

Dynamics

- Players learn to target the spotter first, and the front-most Pawn during a ceremony.
- Lines of braced Pawns create zones the player must flank around rather than charge.
- Players bait Pawns into bracing, then step out of the line to make them waste thrusts.
- The fight escalates when a group shrinks, because losing a Pawn can trigger a promotion.

Aesthetics

- **Challenge:** the tension of facing a coordinated crowd that is weak individually.
- **Discovery:** learning to read formations and find the gaps in them.
- **Narrative:** Alice outwitting a rule-bound army that mirrors her own chessboard story.

Access

- The brace is visible: the Pawn stops, lowers its spear, and winds up, so the player can read when a long thrust is coming.
- When a Pawn spots the player, a visible signal (a flag or horn) shows the player has been seen.
- During Coronation the Pawns gather around the chosen Pawn, and a silhouette of the incoming piece appears above it, so the player knows what is coming and can respond.

8. Schell Lenses

#19 The Lens of the Player

What does the player expect from a lone Pawn, and what do they actually find? In the paper playtest, players found the Pawn nearly blind and easy to kill, and its sideways diagonal attack forced it to face away from the player, which felt awkward. I changed the attack to a forward jab and a braced thrust, and moved the unit's strength from the individual to the group.

#28 The Lens of the State Machine

Are the Pawn's states and transitions clear to the player? The original promotion meter was a hidden, continuous state that the player could not track across many Pawns. I replaced it with discrete, visible states (March, Brace, Coronation) that the player can recognize on sight.

#39 The Lens of Meaningful Choices

Does the player face a real decision when meeting a Pawn group? Originally the only choice was to kill one nearly blind Pawn. Now the player chooses between killing the spotter, flanking a braced line, interrupting a Coronation, or retreating, and each choice has a cost.

#37 The Lens of Fairness

Is the random promotion result fair to the player? Randomness alone could feel arbitrary. I made the result visible during the ceremony through a silhouette, limited promotion to once per group, and made promoted forms temporary and weaker than the real units.

#30 The Lens of Emergence

What behavior comes from simple rules when many Pawns are present? Phalanx reach and Shared Sight create lines, choke points, and flanking opportunities without extra scripted behavior. This led me to add group rules instead of more per-Pawn abilities.

9. Senet X Carry-Forward

In Nahl, our Senet and Hive hybrid, the biggest risk we flagged was that the die could decide more than the player, and playtesters described choosing their strategy around whatever number came up. Our revision was not to remove the dice but to give players a decision about how to use them: two dice, each moving a different piece of the player's choice. The lesson I carried forward is that randomness feels fair when the player can see it and still act on it. The Pawn's promotion is random (Knight, Rook, or Queen), but the result is revealed at the start of the ceremony through a silhouette, and the player can still respond by killing the chosen Pawn, repositioning, or retreating, so chance shapes the fight without taking choices away.

10. Unity Behavior Proof (Digital)

Item	Value
Repository	github.com/JaeshinJung/VIST-386-Minor-Proj-3
Versions	Unity 6000.3.26f1, Unity Behavior (com.unity.behavior) 1.0.16
Base	Unity Behavior Example sample imported through Package Manager (not the Asset Store demo)
Graph	Assets/Samples/Behavior/1.0.16/Unity Behavior Example/Example Behavior Graph.asset
Video	MP3_Pawn_Behavior_Demo.mp4 (56 s)

What it demonstrates. The Pawn's ENGAGE decision from the Behavior Tree. After On Start (Repeat) and "What should I do?", a **Branch On** node checks the condition **PlayerDistance < 3**. If true, the agent performs the **Jab** outcome (says "JAB! (close range, moving attack)" and holds for 1 second). If false, it performs the **Braced Thrust** outcome (says "BRACE + THRUST! (far range)", picks a target, and navigates to it). The Blackboard holds **PlayerDistance** (float) and **IsBraced** (bool) alongside Self, Target, and Speed. Changing PlayerDistance on the Behavior Agent in Play mode switches the branch, as shown in the video.

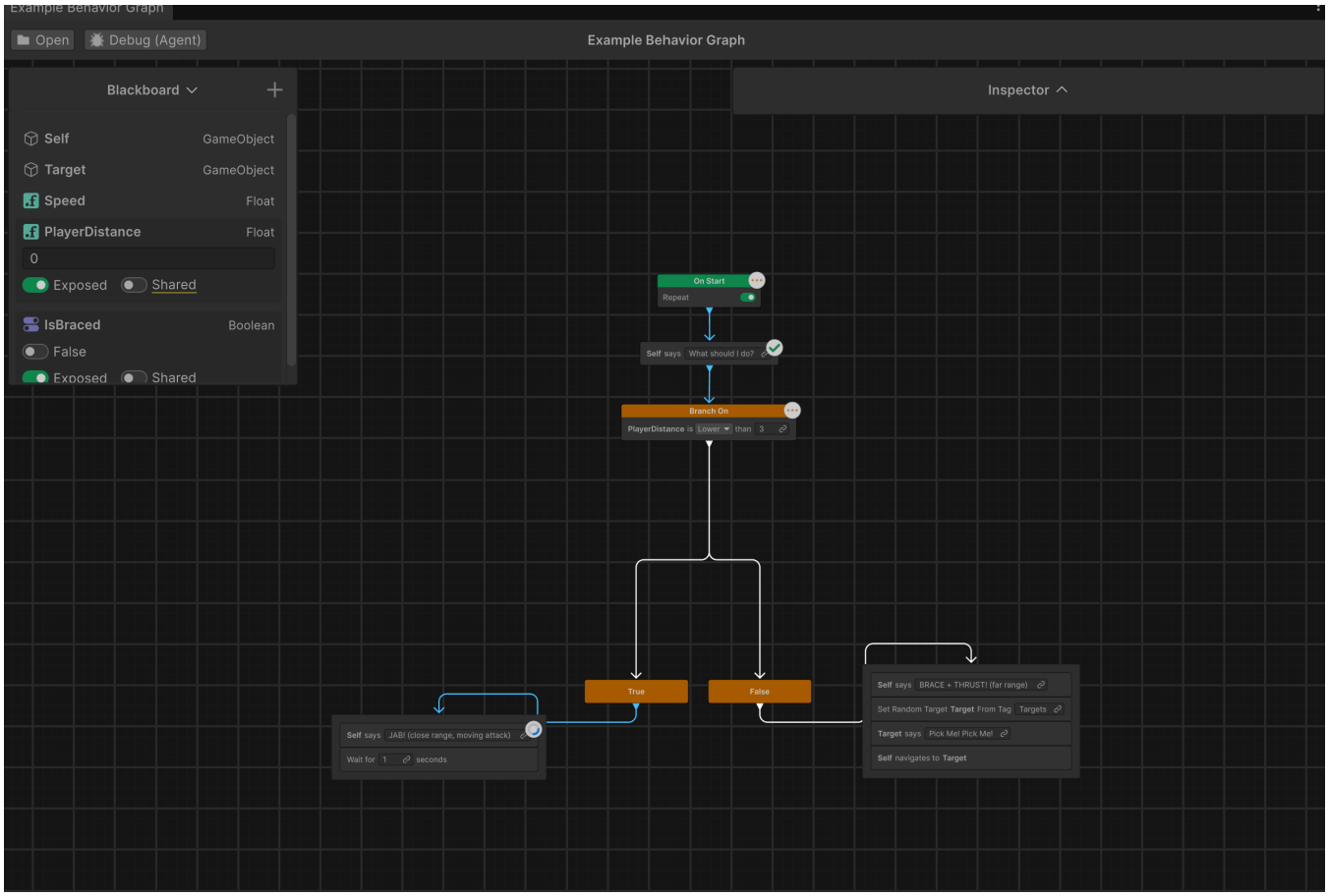


Figure 3. Behavior Graph in Play mode with the debugger attached to Agent. The blue path shows execution through Branch On (PlayerDistance Lower than 3) into the True / Jab branch.